



DATABASE MANAGEMENT SYSTEMS

Construction Material & Site Progress Management System

DBMS Project — Project 44 · Review 1

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Course: Database Management Systems



■ Problem Identification

Construction companies need to manage project, material, contractor, site progress, and billing information efficiently. Disconnected or manual records can lead to material shortages and inaccurate progress payments. A centralized database system is therefore required to integrate these activities and maintain accurate project information.

Manual / Disconnected
Records

Material Shortages &
Inaccurate Progress Payments

Need for a Centralized
Database

■ Scope & Objectives

Scope



Objectives

- Manage projects, sites, contractors and work packages
- Manage materials, suppliers, purchase orders and deliveries
- Track site stock and material issues
- Record work progress and labour teams
- Verify contractor bills and manage payments

Reports Generated

- Material balance
- Work progress
- Supplier performance
- Contractor bills
- Cost variance
- Project status

■ Users & Functional Requirements

Users

Proposed roles, derived from the system's functions

- Project Manager
- Site Engineer
- Procurement Staff
- Site / Store Staff
- Accounts / Billing Staff
- Management

Project & Work

- Manage projects, sites, contractors and work packages
- Manage labour teams and progress entries

Materials & Procurement

- Manage materials and suppliers
- Create purchase orders and record deliveries
- Track site stock and material issues

Billing & Reporting

- Verify contractor bills and record payments
- Generate material balance, work progress, supplier performance, contractor bill, cost variance and project status reports

■ Key Business Rules

■ Material Issue

Cannot exceed available site stock

■ Progress

Must remain between 0% and 100%

■ Contractor Billing

Billed quantity must be within verified work

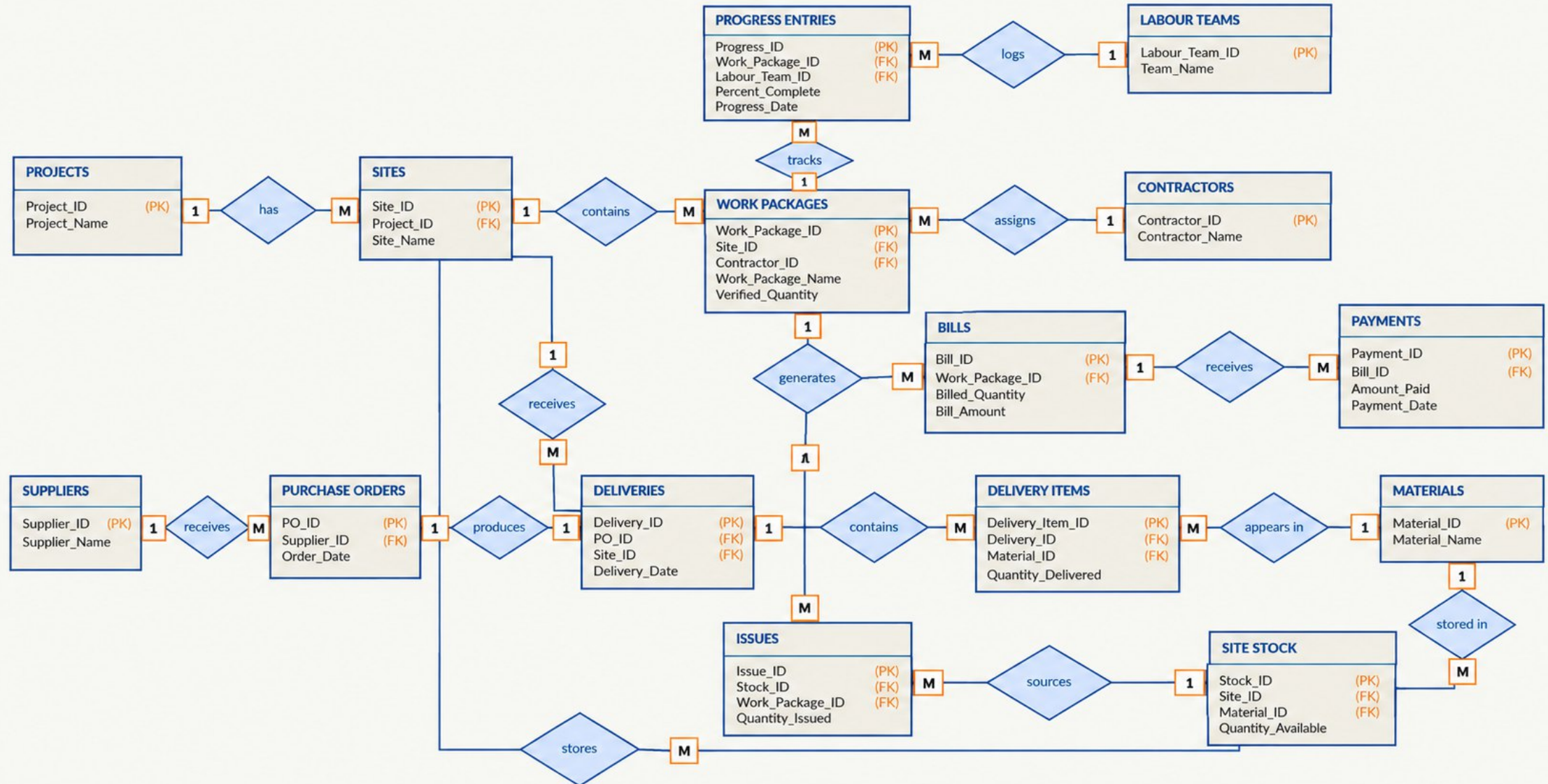
■ Payment

Cannot exceed the approved bill

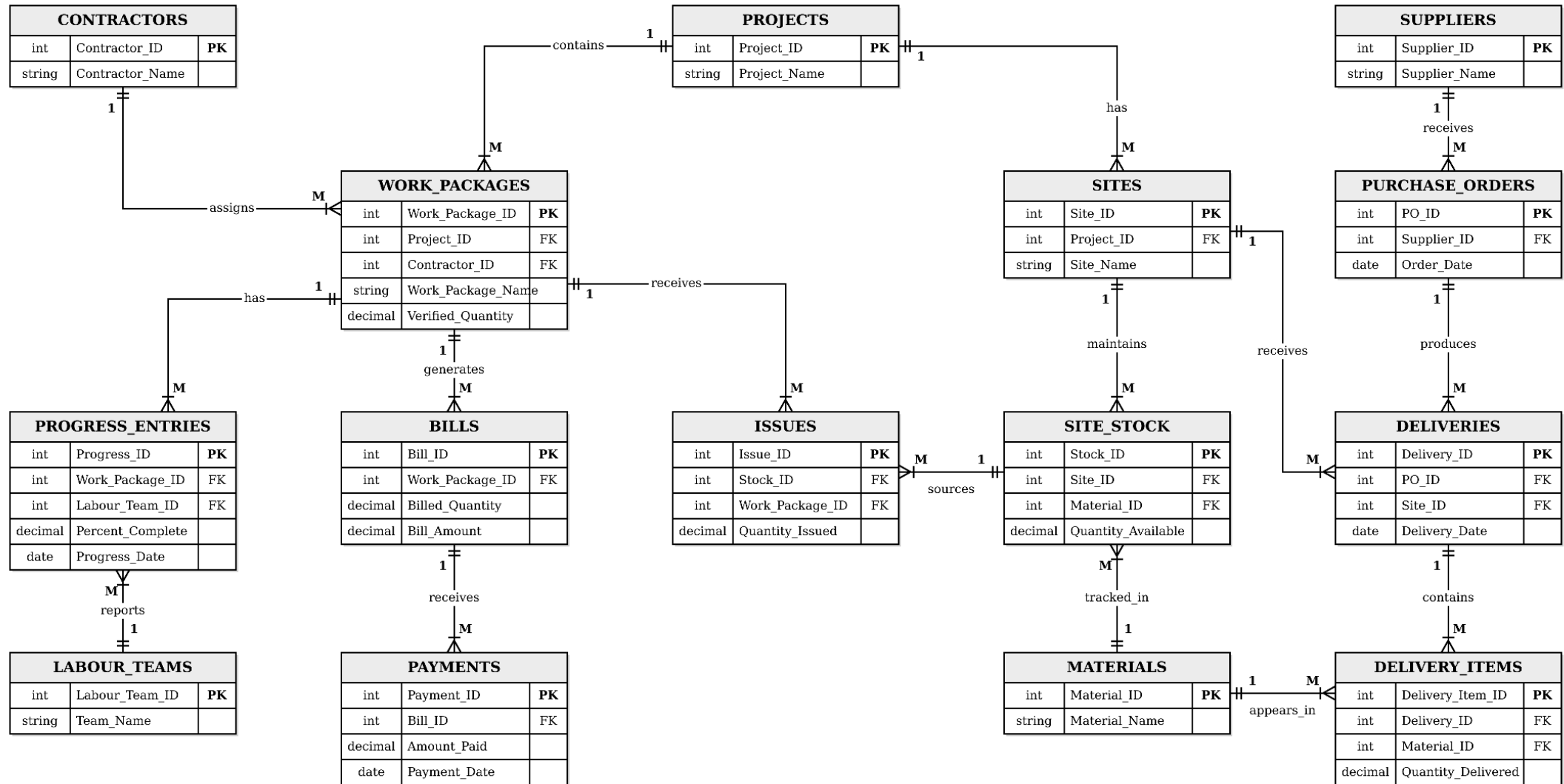
■ Delivery

Delivery dates must be valid

■ Conceptual Entity-Relationship Diagram



Initial Relational Schema



■ Normalization & Functional Dependencies

Functional Dependencies

Project_ID	→ Project_Name
Site_ID	→ Project_ID, Site_Name
Contractor_ID	→ Contractor_Name
Supplier_ID	→ Supplier_Name
Work_Package_ID	→ Project_ID, Contractor_ID, Work_Package_Name, Verified_Quantity
Material_ID	→ Material_Name
PO_ID	→ Supplier_ID, Order_Date
Delivery_ID	→ PO_ID, Site_ID, Delivery_Date
Delivery_Item_ID	→ Delivery_ID, Material_ID, Quantity_Delivered
Stock_ID	→ Site_ID, Material_ID, Quantity_Available
Issue_ID	→ Stock_ID, Work_Package_ID, Quantity_Issued
Labour_Team_ID	→ Team_Name
Progress_ID	→ Work_Package_ID, Labour_Team_ID, Percent_Complete, Progress_Date
Bill_ID	→ Work_Package_ID, Billed_Quantity, Bill_Amount
Payment_ID	→ Bill_ID, Amount_Paid, Payment_Date

1NF

Atomic attributes; no repeating groups.



2NF

No partial dependencies; tables use single-attribute primary keys.



3NF

No transitive dependencies; non-key attributes depend only on their primary key.

The final database design is normalized up to 3NF, reducing redundancy and maintaining data integrity.

■ MySQL 8.0 Implementation & Testing

Database Implementation

- Database: Construction_Site_Management
- 15 tables created
- Primary Key and Foreign Key constraints
- NOT NULL, CHECK and referential integrity constraints
- Sample data inserted across all tables
- Database relationships implemented according to the ERD

Business Rule Testing

- Invalid progress percentage →Rejected
- Issue quantity exceeding available stock →Rejected
- Bill quantity exceeding verified quantity →Rejected
- Invalid delivery date →Rejected
- Payment exceeding bill amount →Rejected

■ SQL Queries, Views & Reports

SQL Queries

- Multi-table JOIN queries
- Aggregate functions
- Subqueries
- Filtering & grouping
- Construction, site, material, progress and billing queries

Views

- Material Balance
- Work Progress
- Supplier Performance
- Contractor Bills
- Cost Variance
- Project Status

Reports

- Material & Stock Report
- Site Progress Report
- Supplier Performance Report
- Contractor Billing & Payment Report
- Project Cost Variance
- Project Status Summary